

Section 3: Simple Linear Regression

STAT 139 Teaching Team

1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See Canvas for the most accurate information!

1.3 Attendance

- Please notify the Teaching Fellow (TF) if your attendance hasn't been taken!

2 Notation and Vocabulary

Idea. As we expand into *higher dimensions*, it is important to keep our notation clear and consistent.

2.1 Parameters, Estimators, and Estimates

Definition 1 (Parameter). A value that specifies a model—typically unknown.

- E.g., let $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(\mu)$, with $\mu = \mathbb{E}[Y_i]$ unknown. Here, μ is a parameter.

Definition 2 (Estimator). A function of the random data as our “best guess” of something unknown.

- E.g., let $\hat{\mu} = \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. Here, $\hat{\mu}$ is an estimator for μ .

Definition 3 (Estimate). A realized value of the estimator once we observe data.

- E.g., with our observations, we can calculate $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$. Here, $\bar{y}$ is an estimate.¹

2.2 What Does S Mean?

- $s_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2$ (i.e., sum of squares with X).
- $s_{yy} = \sum_{i=1}^n (Y_i - \bar{Y})^2$ (i.e., sum of squares with Y).
- $s_{xy} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$ (i.e., sum of squares with X and Y).
- $S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ (i.e., sample variance for X).

¹In this example, $\hat{\mu}$ can refer to either the estimator or estimate, and which of the two it is should be clear from context.

- $S_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ (i.e., sample variance for Y).
- $S_X = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$ (i.e., sample standard deviation for X).
- $S_Y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}$ (i.e., sample standard deviation for Y).

 **Concept Checker 1.** How can we rewrite S_X^2 in terms of s_{xx} ?

Solution

$$S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} s_{xx}.$$

2.3 Moving Up Dimensions

Dimension	Constant	Random
Scalar	Lowercase: c	Uppercase: X
Vector	Lowercase w/ arrow: $\vec{c}$ Lowercase w/ bold: $\mathbf{c}$	Uppercase w/ arrow: $\vec{X}$ Uppercase w/ bold: $\mathbf{X}$
Matrix	Uppercase: C Uppercase w/ bold: $\mathbf{C}$	Uppercase: X Uppercase w/ bold: $\mathbf{X}$

Definition 4 (Vectors and matrices). Unless otherwise noted, $\vec{c}$ refers to a *column* vector:

$$\vec{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

- Thus, the matrix $\mathbf{C}$ can be interpreted as a collection of *column* vectors:

$$\mathbf{C} = (\vec{c}_1 \mid \vec{c}_2 \mid \dots \mid \vec{c}_m) = \begin{pmatrix} c_{1,1} & \dots & c_{1,m} \\ \vdots & \ddots & \vdots \\ c_{n,1} & \dots & c_{n,m} \end{pmatrix}.$$

Definition 5 (Random vector). $\vec{X}$ is a *random vector* if its components are random variables.

- Similarly, $\mathbf{X}$ is a *random matrix* if its components are random variables.²

3 Predictive Regression

Idea. We don't know the future, but statistics allows us to better *understand* and *predict* outcomes using data! We begin the course with *simple linear regression*, which will serve as the foundation for everything else we'll learn.

²Unfortunately, there isn't great notation to differentiate between a constant matrix and a random matrix, but which of the two it is should be clear from context.

3.1 Fundamentals

Definition 6 (Outcome). For observation $i \in \{1, \dots, n\}$, we are interested in predicting/modeling the *outcome* Y_i —a.k.a. the *response*, *output*, and *dependent variable*.³

- This is a *random variable*.
- E.g., we are interested in predicting someone's SAT score.

Definition 7 (Predictor). For observation $i \in \{1, \dots, n\}$, we model Y_i as a function of the *predictor* x_i —a.k.a. the *input*, *explanatory*, and *independent variable*.

- This is a *fixed quantity*. We treat this value as non-random.⁴
- More generally, we can work with p *predictors*: $x_{i,1}, \dots, x_{i,p}$ for each observation i . For simplicity, we begin with $p = 1$.
- E.g., we can model someone's SAT score as a function of their GPA.

Definition 8 (Error). For observation $i \in \{1, \dots, n\}$, let $Y_i = f(x_i) + \varepsilon_i$. We include an *individualized error term* ε_i so that two observations with the same x_i can have different observed values of Y_i .

- This is a *random variable*. In practice, this value is never known.
- We decompose Y_i into a *systematic part*—i.e., $f(x_i)$ —and a *random part*—i.e., ε_i .
- E.g., Alice and Bob both have a 4.0 GPA, but our model is flexible enough to acknowledge that they might not have the same SAT score. However, they will have the same *predicted* SAT score!

Definition 9 (Linear regression). We assume our model is *linear in the parameters*:

$$Y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j} + \varepsilon_i.$$

- When $p = 1$, $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.
- The form $f(x_i)$ is very general and thus hard to estimate/interpret. Thus, the linearity assumption greatly simplifies our problem.⁵

 **Concept Checker 2.** Which of the following models are linear in their parameters?

1. $Y_i = \beta x_i + \varepsilon_i$
2. $Y_i = \beta_0 + \beta_1 x_i^{\beta_2} + \varepsilon_i$
3. $Y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2} + \varepsilon_i$
4. $Y_i = \beta \sin(x_i) + \varepsilon_i$

³In *regression*, the outcome is continuous. We will learn about *classification* later in the course.

⁴We'll often denote the predictor with a lowercase x to indicate that it is a *crystallized* value. Sometimes, it is denoted with a capital X to indicate that it is a *random variable*, but then expectation is conditional on $X = x$. E.g., if $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, then $\mathbb{E}[Y_i | X_i = x_i] = \beta_0 + \beta_1 x_i$. Unless otherwise noted, we work with a sample space conditional on $X = x$.

⁵Other models exist, such as generalized additive models (GAMs), but these lose much of the interpretability that makes linear regression so powerful.

Solution

All models except 2 are linear in their parameters. Even though $x_{i,1}x_{i,2}$ and $\sin(x_i)$ are not linear functions, these are functions of the predictors. We are concerned about whether the model is linear in the parameters.

3.2 Simple Linear Regression

Definition 10 (Simple linear regression). With $p = 1$,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

- We have an *intercept* parameter and a *slope* parameter with one *predictor*.
- In *simple linear regression*, $p = 1$, but in *multiple linear regression*, $p > 1$.

Definition 11 (Population regression function). Assuming that $\mathbb{E}[\varepsilon_i] = 0$,

$$\mu_{Y_i} = \mathbb{E}[Y_i] = \beta_0 + \beta_1 x_i.$$

Definition 12 (Intercept term). β_0 is the *expected* outcome when $x_i = 0$.

Definition 13 (Slope term). β_1 is the *expected* change in Y_i associated with a one-unit increase in x_i .⁶

Definition 14 (Variance term). σ_i^2 is the *variance* in the error and thus the outcome (i.e., $\sigma_i^2 = \text{Var}[\varepsilon_i] = \text{Var}[Y_i]$).

 **Concept Checker 3.** How many unknown parameters are there in simple linear regression? Assume constant variance (i.e., homoskedasticity).

Solution

There are 3: β_0 , β_1 , and σ^2 . Importantly, if β_0 and β_1 were known, then ε_i would be deterministic once Y_i crystallizes (but σ^2 would still be unknown).

4 Inference

Idea. A common theme in statistics is the presence of *unknown values*—e.g., the intercept and slope terms. We can *infer* the best estimates from the data, but we must consider that our observed data are random.

4.1 Assumptions

Definition 15 (Existence). $\text{Var}[\varepsilon_i] < \infty$ for all $i \in \{1, \dots, n\}$.

- I.e., the variance is finite.

⁶Importantly, this is not necessarily causal, but this is sometimes referred to as the “effect” of x_i on Y_i .

Definition 16 (Linearity). $\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_i$, which implies that $\mathbb{E}[\varepsilon_i] = 0$ for all $i \in \{1, \dots, n\}$.

- I.e., the population regression function holds.

Definition 17 (Independence). $\varepsilon_i \perp\!\!\!\perp \varepsilon_j$ for all $i \neq j$.

- I.e., the errors and thus the outcomes are independent.

Definition 18 (Homoskedasticity). $\text{Var}[\varepsilon_i] = \sigma^2$ for all $i \in \{1, \dots, n\}$.

- I.e., the variance is the same for all errors and thus outcomes.

Definition 19 (Normality). $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ for all $i \in \{1, \dots, n\}$.

- I.e., the errors and thus the outcomes are Normal.

Definition 20 (ELIHN). Together, the five main assumptions can be summarized as $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, with $\sigma^2 < \infty$.

- “Eli H. Normal” is a good way to remember this.
- Importantly, the assumptions are ordered from *most to least important*. Much of this course will be dedicated to breaking down these assumptions!
- Related, the fifth assumption of Normality is *not* needed for OLS point estimation but is often used for exact finite-sample inference.

4.2 Prediction

Definition 21 (Predicted outcome). Suppose that we have $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\sigma}^2$ as estimates from our data. Naturally, we calculate our *predicted outcome* as

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

Definition 22 (Residual).

$$e_i = Y_i - \hat{Y}_i$$

is the *residual* (i.e., the difference in *predicted* and *observed* outcomes).

- This is *observed* once Y_i crystallizes.
- Equivalently, this can be denoted as $\hat{\varepsilon}_i$ to emphasize that it’s an estimator for ε_i .

 **Concept Checker 4.** We use simple linear regression to model someone’s SAT score (i.e., $Y_i \in [400, 1600]$) as a function of their GPA (i.e., $x_i \in [0.0, 4.0]$). From the data, we calculate $\hat{\beta}_0 = 400$ and $\hat{\beta}_1 = 300$. What is the predicted outcome for Alice if she has a 4.0 GPA? If she actually scored a 1500, what is her residual?

Solution

Our model is $\hat{Y}_i = 400 + 300x_i$, so $\hat{Y}_{\text{Alice}} = 400 + 300(4.0) = 1600$. Thus, $e_{\text{Alice}} = 1500 - 1600 = -100$.

4.3 Ordinary Least-Squares (OLS) Estimation

Definition 23 (Objective function). What we want to *minimize* to form our estimators.

- In this course, we'll often use *sum of squares errors* (SSE).⁷
- The *normal equations* are the derivatives set to 0: $\frac{\partial}{\partial \beta_0} \text{SSE} = 0$, $\frac{\partial}{\partial \beta_1} \text{SSE} = 0$.

Definition 24 (Sum of squares errors (SSE)).

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

- Since $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, we can rewrite this as

$$\text{SSE} = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2.$$

 **Concept Checker 5.** Is SSE theoretical (i.e., unobservable)?

Solution

No! It can be calculated with our data (once Y_i crystallizes), and it's a good way to measure how good our model is using the data at hand.

Definition 25 (OLS estimators). $(\hat{\beta}_{0,\text{OLS}}, \hat{\beta}_{1,\text{OLS}}) = \arg \min_{(\beta_0, \beta_1) \in \mathbb{R}^2} (\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2)$.

- I.e., our estimators are the values for β_0 and β_1 that *minimize* SSE:

$$\hat{\beta}_{1,\text{OLS}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

and

$$\hat{\beta}_{0,\text{OLS}} = \bar{Y} - \hat{\beta}_{1,\text{OLS}} \bar{x}.$$
⁸

Definition 26 (Variance estimator).

$$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

- Equivalently,

$$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n \left(Y_i - (\hat{\beta}_{0,\text{OLS}} + \hat{\beta}_{1,\text{OLS}} x_i) \right)^2 = \frac{\text{SSE}}{\text{df}}.$$

- Intuitively, we lose 2 *degrees of freedom* since we use the data to calculate $\hat{\beta}_{0,\text{OLS}}$ and $\hat{\beta}_{1,\text{OLS}}$.⁹

⁷This term is confusing because it uses *residuals*—which are observable—not *errors*—which are unobservable. Equivalently, this is sometimes denoted as *residual sum of squares* (RSS).

⁸See Appendix A.1.

⁹E.g., if I told you that we had 3 observations with $y_1 = 1$, $y_2 = 1$, and $\bar{y} = 1$, then you'd be able to determine that $y_3 = 1$, so there's no randomness (i.e., no freedom to vary) in that last observation!

 **Concept Checker 6.** In OLS simple linear regression, what two important points must the regression line always pass through?

Solution

First, $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, so if $x_i = 0$, then $\hat{Y}_i = \hat{\beta}_0$. Therefore, one point is $(0, \hat{\beta}_0)$. Additionally, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, so $\hat{Y}_i = \bar{Y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i = \bar{Y} - \hat{\beta}_1 (\bar{x} - x_i)$. Thus, if $x_i = \bar{x}$, then $\hat{Y}_i = \bar{Y}$. Therefore, another point is $(\bar{x}, \bar{Y})$.

4.4 Prediction Error

Definition 27 (Sum of squares errors (SSE)).

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

- This is a measure of total deviation from using our *predicted outcome* $\hat{Y}_i$.

Definition 28 (Sum of squares regression (SSR)).

$$\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2.$$

- This is a measure of how much “better” it is to use $\hat{Y}_i$ than $\bar{Y}$ (i.e., how much error we “explain away”).

Definition 29 (Sum of squares total (SST)).

$$\text{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

- This is a measure of total deviation from using $\bar{Y}$ as a simple estimator.
- Equivalently,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Definition 30 (R^2). R^2 is the proportion of the *total variability* in Y that is explained by the *regression* of Y on X (i.e., when using X to predict Y):

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

- Equivalently,

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

- Let r_{XY} be the Pearson correlation coefficient between X and Y .¹⁰ In *simple linear regression*, $R^2 = (r_{XY})^2$.

¹⁰This is an estimator for $\rho = \text{Corr}[X, Y]$.

5 Exercises

 **Problem 1.** In linear regression, there are two important identities often used.

- (a) Show that $\sum_{i=1}^n (x_i - \bar{x}) = 0$.¹¹
 (b) Show that $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$.¹²

Solution

(a) First, we'll show

$$\begin{aligned}
 \sum_{i=1}^n (x_i - \bar{x}) &= \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} && \text{by linearity} \\
 &= n \left(\frac{1}{n} \sum_{i=1}^n x_i \right) - \sum_{i=1}^n \bar{x} && \text{since } n \times \frac{1}{n} = 1 \\
 &= n\bar{x} - n\bar{x} && \text{since } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \\
 &= 0 && \text{by simplifying}
 \end{aligned}$$

(b)

$$\begin{aligned}
 \sum_{i=1}^n (x_i - \bar{x})y_i &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y} + \bar{y}) && \text{since } -\bar{y} + \bar{y} = 0 \\
 &= \sum_{i=1}^n ((x_i - \bar{x})(y_i - \bar{y}) + (x_i - \bar{x})\bar{y}) && \text{by distributing} \\
 &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \bar{y} \sum_{i=1}^n (x_i - \bar{x}) && \text{by linearity} \\
 &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \bar{y} \times 0 && \text{since } \sum_{i=1}^n (x_i - \bar{x}) = 0 \\
 &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) && \text{by simplifying}
 \end{aligned}$$

 **Problem 2.** Consider the following linear model:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where

$$\varepsilon_i \stackrel{\text{ind.}}{\sim} \mathcal{N}(0, \sigma^2 w_i^{-1})$$

¹¹Hint: $\sum_{i=1}^n (a + b) = \sum_{i=1}^n a + \sum_{i=1}^n b$.

¹²Hint: A useful trick is to add and subtract by the same number (thereby adding 0). This is also true for multiplying and dividing by the same number.

for all $i \in \{1, \dots, n\}$.

The “weighted least squares” (WLS) estimators are those that minimize

$$Q(\vec{\beta}) = \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i)^2.$$

In what follows, it will help to define

$$\bar{Y}_w = \frac{\sum_{i=1}^n w_i Y_i}{\sum_{i=1}^n w_i}, \quad \bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}.^{13}$$

(a) Derive the WLS estimators of β_0 and β_1 .

(b) Derive the variance of $\hat{\beta}_{1,\text{WLS}}$. It might help to first show:

$$\sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) = \sum_{i=1}^n w_i (x_i - \bar{x}_w) Y_i.$$

Solution

(a) To find where $Q(\vec{\beta}) = \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i)^2$ is minimized, we can set the partial derivatives (with respect to β_0 and β_1) equal to 0 and solve the system of equations—i.e., $\frac{\partial}{\partial \beta_0} Q(\vec{\beta}) = 0$, $\frac{\partial}{\partial \beta_1} Q(\vec{\beta}) = 0$.

$$\begin{aligned} \frac{\partial}{\partial \beta_0} \left(\sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i)^2 \right) &= \sum_{i=1}^n \frac{\partial}{\partial \beta_0} (w_i (Y_i - \beta_0 - \beta_1 x_i)^2) && \text{by linearity} \\ &= \sum_{i=1}^n 2w_i (Y_i - \beta_0 - \beta_1 x_i)(-1) && \text{by chain rule} \\ &= -2 \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i) && \text{by simplifying} \end{aligned}$$

$$\begin{aligned} 0 = -2 \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i) &\implies 0 = \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i) && \text{by multiplying} \\ &\implies 0 = \sum_{i=1}^n w_i Y_i - \beta_0 \sum_{i=1}^n w_i - \beta_1 \sum_{i=1}^n w_i x_i && \text{by linearity} \\ &\implies \beta_0 = \frac{\sum_{i=1}^n w_i Y_i - \beta_1 \sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} && \text{by solving for } \beta_0 \\ &\implies \beta_0 = \bar{Y}_w - \beta_1 \bar{x}_w && \text{by simplifying} \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 x_i)^2 \right) &= \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n w_i (Y_i - (\bar{Y}_w - \beta_1 \bar{x}_w) - \beta_1 x_i)^2 \right) && \text{since } \beta_0 = \bar{Y}_w - \beta_1 \bar{x}_w \\ &= \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n w_i (Y_i - \bar{Y}_w - \beta_1 (x_i - \bar{x}_w))^2 \right) && \text{by distributing} \\ &= \sum_{i=1}^n \frac{\partial}{\partial \beta_1} w_i (Y_i - \bar{Y}_w - \beta_1 (x_i - \bar{x}_w))^2 && \text{by linearity} \\ &= \sum_{i=1}^n w_i (2(Y_i - \bar{Y}_w - \beta_1 (x_i - \bar{x}_w))(-1)(x_i - \bar{x}_w)) && \text{by chain rule} \\ &= -2 \sum_{i=1}^n w_i ((Y_i - \bar{Y}_w)(x_i - \bar{x}_w) - \beta_1 (x_i - \bar{x}_w)^2) && \text{by simplifying} \end{aligned}$$

¹³Inspired by Problem 4 in “Stat 139 Midterm Exam 1, Fall 2025” by James Xenakis.

$$\begin{aligned}
0 &= -2 \sum_{i=1}^n w_i ((Y_i - \bar{Y}_w)(x_i - \bar{x}_w) - \beta_1(x_i - \bar{x}_w)^2) \implies 0 = \sum_{i=1}^n w_i ((Y_i - \bar{Y}_w)(x_i - \bar{x}_w) - \beta_1(x_i - \bar{x}_w)^2) && \text{by mult.} \\
&\implies 0 = \sum_{i=1}^n w_i (Y_i - \bar{Y}_w)(x_i - \bar{x}_w) - \beta_1 \sum_{i=1}^n w_i (x_i - \bar{x}_w)^2 && \text{by lin.} \\
&\implies \beta_1 = \frac{\sum_{i=1}^n w_i (Y_i - \bar{Y}_w)(x_i - \bar{x}_w)}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2} && \text{by alg.}
\end{aligned}$$

Thus, $\hat{\beta}_{0,\text{WLS}} = \bar{Y}_w - \hat{\beta}_{1,\text{WLS}}\bar{x}_w$ and $\hat{\beta}_{1,\text{WLS}} = \frac{\sum_{i=1}^n w_i (Y_i - \bar{Y}_w)(x_i - \bar{x}_w)}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}$.

(b) First, let's show $\sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) = \sum_{i=1}^n w_i (x_i - \bar{x}_w)Y_i$.

$$\begin{aligned}
\sum_{i=1}^n w_i (x_i - \bar{x}_w)Y_i &= \sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w + \bar{Y}_w) && \text{since } -\bar{Y}_w + \bar{Y}_w = 0 \\
&= \sum_{i=1}^n (w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) + w_i (x_i - \bar{x}_w)\bar{Y}_w) && \text{by distributing} \\
&= \sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) + \bar{Y}_w \sum_{i=1}^n w_i (x_i - \bar{x}_w) && \text{by linearity} \\
&= \sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) + \bar{Y}_w \times 0 && \text{since } \sum_{i=1}^n w_i (x_i - \bar{x}_w) = 0 \\
&= \sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w) && \text{by simplifying}
\end{aligned}$$

Next, let's find $\text{Var}[\hat{\beta}_{1,\text{WLS}}]$.

$$\begin{aligned}
\text{Var}[\hat{\beta}_{1,\text{WLS}}] &= \text{Var}\left[\frac{\sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w)}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right] && \text{by substituting} \\
&= \left(\frac{1}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right)^2 \text{Var}\left[\sum_{i=1}^n w_i (x_i - \bar{x}_w)(Y_i - \bar{Y}_w)\right] && \text{by bilinearity} \\
&= \left(\frac{1}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right)^2 \text{Var}\left[\sum_{i=1}^n w_i (x_i - \bar{x}_w)Y_i\right] && \text{by identity shown above} \\
&= \left(\frac{1}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right)^2 \sum_{i=1}^n \text{Var}[w_i (x_i - \bar{x}_w)Y_i] && \text{by independence} \\
&= \left(\frac{1}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right)^2 \sum_{i=1}^n (w_i (x_i - \bar{x}_w))^2 \text{Var}[Y_i] && \text{by bilinearity} \\
&= \left(\frac{1}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}\right)^2 \sum_{i=1}^n (w_i (x_i - \bar{x}_w))^2 (\sigma^2 w_i^{-1}) && \text{since } \text{Var}[\varepsilon_i] = \text{Var}[Y_i] = \sigma^2 w_i^{-1} \\
&= \frac{\sigma^2}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2} && \text{by simplifying}
\end{aligned}$$

A Proofs

A.1 OLS Estimators Minimize SSE

Proof. Let $\text{SSE} = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$. We want to show that $\hat{\beta}_{1,\text{OLS}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_{0,\text{OLS}} = \bar{Y} - \hat{\beta}_{1,\text{OLS}}\bar{x}$ are the arguments that minimize SSE.

To find where SSE is minimized, we can set the partial derivatives (with respect to β_0 and β_1) equal to 0 and solve the system of equations—i.e., $\frac{\partial}{\partial \beta_0} \text{SSE} = 0$, $\frac{\partial}{\partial \beta_1} \text{SSE} = 0$.

$$\begin{aligned}
\frac{\partial}{\partial \beta_0} \left(\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2 \right) &= \frac{\partial}{\partial \beta_0} \left(\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 \right) && \text{by distributing} \\
&= \sum_{i=1}^n \frac{\partial}{\partial \beta_0} (Y_i - \beta_0 - \beta_1 x_i)^2 && \text{by linearity} \\
&= \sum_{i=1}^n 2(Y_i - \beta_0 - \beta_1 x_i)(-1) && \text{by chain rule} \\
&= -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) && \text{by simplifying}
\end{aligned}$$

$$\begin{aligned}
0 = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) &\implies 0 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) && \text{by multiplying} \\
&\implies 0 = \sum_{i=1}^n Y_i - n\beta_0 - \beta_1 \sum_{i=1}^n x_i && \text{by linearity} \\
&\implies \beta_0 = \bar{Y} - \beta_1 \bar{x} && \text{by solving for } \beta_0
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2 \right) &= \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n (Y_i - (\bar{Y} - \beta_1 \bar{x} + \beta_1 x_i))^2 \right) && \text{since } \beta_0 = \bar{Y} - \beta_1 \bar{x} \\
&= \frac{\partial}{\partial \beta_1} \left(\sum_{i=1}^n (Y_i - \bar{Y} - \beta_1(x_i - \bar{x}))^2 \right) && \text{by distributing} \\
&= \sum_{i=1}^n \frac{\partial}{\partial \beta_1} (Y_i - \bar{Y} - \beta_1(x_i - \bar{x}))^2 && \text{by linearity} \\
&= \sum_{i=1}^n 2(Y_i - \bar{Y} - \beta_1(x_i - \bar{x}))(-1)(x_i - \bar{x}) && \text{by chain rule} \\
&= -2 \sum_{i=1}^n ((Y_i - \bar{Y})(x_i - \bar{x}) - \beta_1(x_i - \bar{x})^2) && \text{by simplifying}
\end{aligned}$$

$$\begin{aligned}
0 = -2 \sum_{i=1}^n ((Y_i - \bar{Y})(x_i - \bar{x}) - \beta_1(x_i - \bar{x})^2) &\implies 0 = \sum_{i=1}^n ((Y_i - \bar{Y})(x_i - \bar{x}) - \beta_1(x_i - \bar{x})^2) && \text{by multiplying} \\
&\implies 0 = \sum_{i=1}^n (Y_i - \bar{Y})(x_i - \bar{x}) - \beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 && \text{by linearity} \\
&\implies \beta_1 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} && \text{by algebra}
\end{aligned}$$

It can be shown that these critical points are global minima via the second derivative test.

Thus, $\hat{\beta}_{0,OLS} = \bar{Y} - \hat{\beta}_{1,OLS}\bar{x}$ and $\hat{\beta}_{1,OLS} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$. $\square$